KEO INTERNATIONAL CONSULTANTS

Thermal Load Calculation Report

Project: Al Zahra Commercial Tower, JLT
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COOLING LOAD SUMMARY

Floor	Area (sqft)	W/sqft	Load (kW)	Load (TR)
Basement (Parking)	3,200	15	4.8	1.4
Podium Parking	3,200	15	4.8	1.4
Ground Floor (Lobby)	3,200	120	38.4	10.9
1st Floor (Office)	3,200	110	35.2	10.0
2nd Floor (Office)	3,200	110	35.2	10.0
3rd – 17th Floor (×15)	48,000	110	528.0	150.1
18th Floor (Penthouse)	3,200	100	32.0	9.1
Roof (AHU/Plant)	—	—	—	—
TOTAL	72,000	—	678.4	192.9

HVAC SYSTEM BREAKDOWN

Total Calculated Load	678.4 kW (192.9 TR)
FAHU Load (Fresh Air)	85.2 kW (24.2 TR)
AC Unit Load (Cooling)	593.2 kW (168.7 TR)
Recommended System	VRF System (Variable Refrigerant Flow)
Safety Factor	10% applied
Design Load (with SF)	746.2 kW (212.2 TR)
Diversity Factor	0.85
Peak Demand	634.3 kW (180.3 TR)

EQUIPMENT SCHEDULE — VRF OUTDOOR UNITS

VRF Outdoor Units	12 × 16 HP units (Daikin VRV-X or equivalent)
Total Capacity	192 HP = 192.9 TR
FAHU Units	4 × 6.0 TR heat recovery type
Exhaust Fans	18 × inline duct fans (toilets & pantries)
Smoke Extract	2 × axial fans (basement parking)
FCU for Lobby	3 × ceiling cassette units

DESIGN CONDITIONS

Outdoor Temperature	46°C DB / 30°C WB (Dubai summer peak)

Indoor Temperature

23°C ± 1°C (offices), 24°C (lobby)

Relative Humidity

50% ± 5%

Fresh Air Rate

8.5 L/s per person (ASHRAE 62.1)

Occupancy

1 person per 10 sqm (offices)